Madhav Donthula

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Education

University of California, Berkeley — Berkeley, CA

Bachelor of Arts in Data Science | Regents' and Chancellor's Scholar — *May 2029*

- **Relevant Coursework**: CS61A, CS61B, CS188, DATAC100, DATAC101, DATAC104, ENGIN 183C, MATH54

Experience

Amazon Web Services (AWS) — June 2026 – August 2026

Software Development Engineer Intern, AWS Lambda — *Seattle, WA*

- Enhanced agent output for the AWS SAM CLI, the command line tool for deploying **Lambda** functions, by monitoring agent interactions across 40+ runs, cutting tokens **22%** and tool calls **60%** per command
- Built **telemetry** classifying agent vs. human SAM CLI runs, turning agent usage into a measurable segment
- Shipped a structured **JSON** output mode for agents while preserving human-readable output for existing users
- Standardized SAM CLI error contracts so agents could recover from failures without re-running commands

Ensure — May 2026 – August 2026

Co-Founder and CTO | Incubator: SkyDeck Pad-13 Batch 22 — *Berkeley, CA*

- Led **product discovery** for a healthcare price transparency startup, interviewing **30+** clinicians, billing leads, and payer mentors to validate the problem space and scope a pricing engine spanning **500+** procedure codes
- Converted interview insights into a ranked roadmap, mapping 5 features to ship an **MVP** in 6 weeks
- Pitched to **15+** mentors and payer contacts across SkyDeck's Batch 22, converting objections into roadmap changes
- Built the pricing engine computing out-of-pocket costs from payer negotiated rates, deductibles, and OOP caps

Synopsys — December 2025 – May 2026

Software & ML Systems Engineer (Contract) | C/C++, Django, Redis, Jest, Git — *Remote*

- Engineered a C++ and Django benchmark visualization dashboard for chip design engineers, building parameterized REST API endpoints and WebSocket feeds to surface real-time simulation metrics across **50+** hardware benchmarks
- Layered **Redis** caching onto hot API routes, dropping repeat latency by **68%** across **10,000+** chip design simulations
- Trained gradient-boosted models to forecast chip power, cutting **2+ simulation hours** needed for design sign-off

Projects

Guardian | *Python, Django, Vue, TypeScript, PostgreSQL, AWS, Docker*

- Won **1st Place** at Cal Hacks **Bright Data** track by architecting and shipping a real-time fraud detection platform
- Deployed an iOS push notification system via APNs and Python, dispatching fraud alerts within **200ms** of detection
- Containerized Django backend and PostgreSQL database with **Docker**, standardizing dev environment

Code Linguo | *AI Speech Tutoring Platform* | *Python, Django, React, TypeScript, Node.js, Git*

- Architected a full-stack multilingual AI speech tutoring platform serving **500+ active learners**, engineering REST API endpoints and async Whisper pipelines that processed 40,000+ audio samples with real-time GPT-4 NPS scoring
- Implemented an auto-assignment grading engine using **Celery** and pytest, integrated into curricula for **200+** students
- Deployed multi-service platform on **Kubernetes**, scaling Whisper pods horizontally to handle peak audio workloads

Skills & Interests

Languages & Frameworks: Python, Java, C++, TypeScript, JavaScript, SQL, React, Node.js, FastAPI, Django, Vue

Product: User interviews, roadmapping, prioritization, PRDs, Figma, Lovable, A/B testing, stakeholder communication

Interests: Hiking, spikeball, pickleball, cooking, watching psychological thrillers, swimming, working out, collecting colognes